

Project Export

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1. Automated Resume Analyzer for Job Portals

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Summary

This project involves building an automated system to parse resumes, extract key information, and identify relevant skills and experiences. Certification confirms expertise in Natural Language Processing (NLP), information extraction, and data structuring from unstructured text.

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Project Overview

Recruitment processes are often bottlenecked by the sheer volume of applications. A single job posting can attract hundreds of resumes, making manual screening impossible. This project aims to build the core engine of an Applicant Tracking System (ATS)—a tool that automatically reads resumes, understands the content, and structures the data for easy filtering.

As an NLP Intern, you will build a sophisticated parser capable of ingesting resumes in various formats (PDF, DOCX) and converting them into a standardized JSON format. This project is a deep dive into unstructured text processing, Named Entity Recognition (NER), and pattern matching.

Key Objectives & Workflow

1. Document Ingestion & Text Extraction

- Format Handling: Resumes come in different file types. You must implement robust file reading capabilities using libraries like pdfminer.six (for PDFs) and python-docx (for Word documents).
- Text Cleaning: Raw extraction often includes noise (headers, footers, page numbers, special characters). You must write cleaning pipelines to normalize whitespace, remove non-ascii characters, and handle bullet points.

2. Section Segmentation

- Structural Analysis: Before extracting details, the system must identify and split the text into logical sections: Contact Info, Summary, Experience, Education, Skills, and Projects.
- Heuristics: Use keyword-based heuristics (e.g., detecting lines containing "Experience" or "Work History" in bold/caps) to segment the document accurately.

3. Information Extraction (The Core NLP Task)

- Contact Details: Use complex Regular Expressions (RegEx) to reliably extract Email Addresses, Phone Numbers, and LinkedIn/Portfolio URLs.

- Named Entity Recognition (NER): Train or fine-tune an NER model (using spaCy or HuggingFace Transformers) to detect Candidate Names, University Names, and Degree Titles.
- Skill Extraction: mapping text against a predefined database of skills (e.g., "Python", "Machine Learning", "Project Management"). You must handle variations (e.g., "React" vs "React.js").

4. Data Structuring

- The final output must be a clean, nested JSON object. For example, the 'Experience' section should be a list of objects, each containing Company Name, Job Title, Dates, and Description.

Prerequisites

To successfully complete this project, you should possess the following skills:

- Programming: Intermediate to Advanced Python.
- Libraries: spaCy, NLTK, pdfminer.six, python-docx, pandas.
- Concepts: Regular Expressions (Regex), Named Entity Recognition (NER), Text Preprocessing (Tokenization, Lemmatization), JSON Data Structures.

Deliverables

1. The Parser Engine

- A Python module (parser.py) containing a class ResumeParser. It should have a method parsefile(filepath) that returns a dictionary/JSON object.

2. The Skill Knowledge Base

- A CSV or JSON file acting as your "Skill Ontology" (a comprehensive list of technical and soft skills used for matching).

3. API Wrapper (Bonus/Advanced)

- A simple Flask or FastAPI script that allows a user to upload a file via a local web server and receive the parsed JSON response.

4. Project Documentation

- A README.md file explaining how to run the parser, required dependencies, and the logic used for section segmentation.

Project Timeline & Deadlines

- Week 1: Ingestion & Segmentation
 - Goal: text extraction and structural understanding.
 - Task: Build script to read PDF/DOCX. Implement logic to slice the text into specific sections (Education vs Experience).
- Week 2: Regex & Contact Info
 - Goal: High-precision extraction of metadata.
 - Task: Implement Regex for Emails, Phone numbers, and Links. Normalize the data (e.g., format all phone numbers to a standard international format).
- Week 3: Entity & Skill Extraction
 - Goal: Intelligent parsing.
 - Task: Use spaCy to find Names and Organizations. Implement the keyword matching logic to extract skills from the text against your Knowledge Base.
- Week 4: Structuring & Final Polish
 - Goal: Production-ready output.
 - Task: Assemble all components into a single JSON output structure. Test against 10-20 different resume layouts to ensure robustness. Submit final code.